Supermarket Sales Analysis Using Excel and SQL

Introduction

This project analyzes supermarket sales data using Microsoft Excel and SQL to identify business trends and generate meaningful insights. The dataset was cleaned and prepared before analysis to ensure accuracy and consistency. SQL was used to answer important business questions related to sales, customers, payment methods, and product categories. Excel was used to create an interactive dashboard with charts for better data visualization. The findings from this project can help businesses make informed decisions regarding sales performance, customer preferences, and business growth.

Data Cleaning

The following data cleaning steps were performed before analysis:

Removed duplicate records to ensure data accuracy.

Checked for missing or incomplete values.

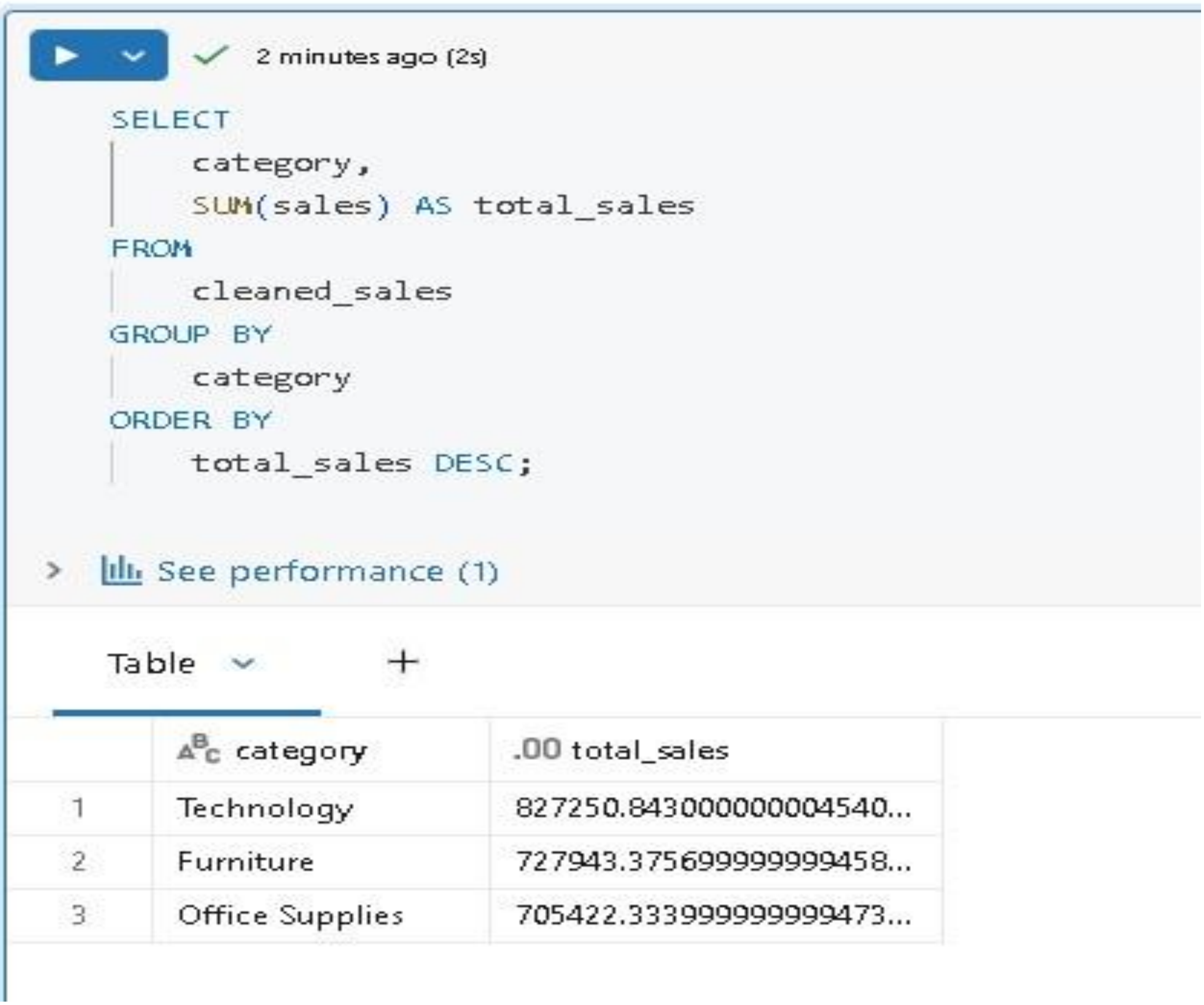
Corrected inconsistent data formats where required.

Standardized column names and data values for consistency.

Verified the dataset to ensure it was ready for SQL analysis and dashboard creation

1. Which product category generates the highest sales?

```
2.  SELECT
3.      category,
4.      SUM(sales) AS total_sales
5.  FROM
6.      cleaned_sales
7.  GROUP BY
8.      category
9.  ORDER BY
10.     total_sales DESC;
```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```
SELECT
    category,
    SUM(sales) AS total_sales
FROM
    cleaned_sales
GROUP BY
    category
ORDER BY
    total_sales DESC;
```

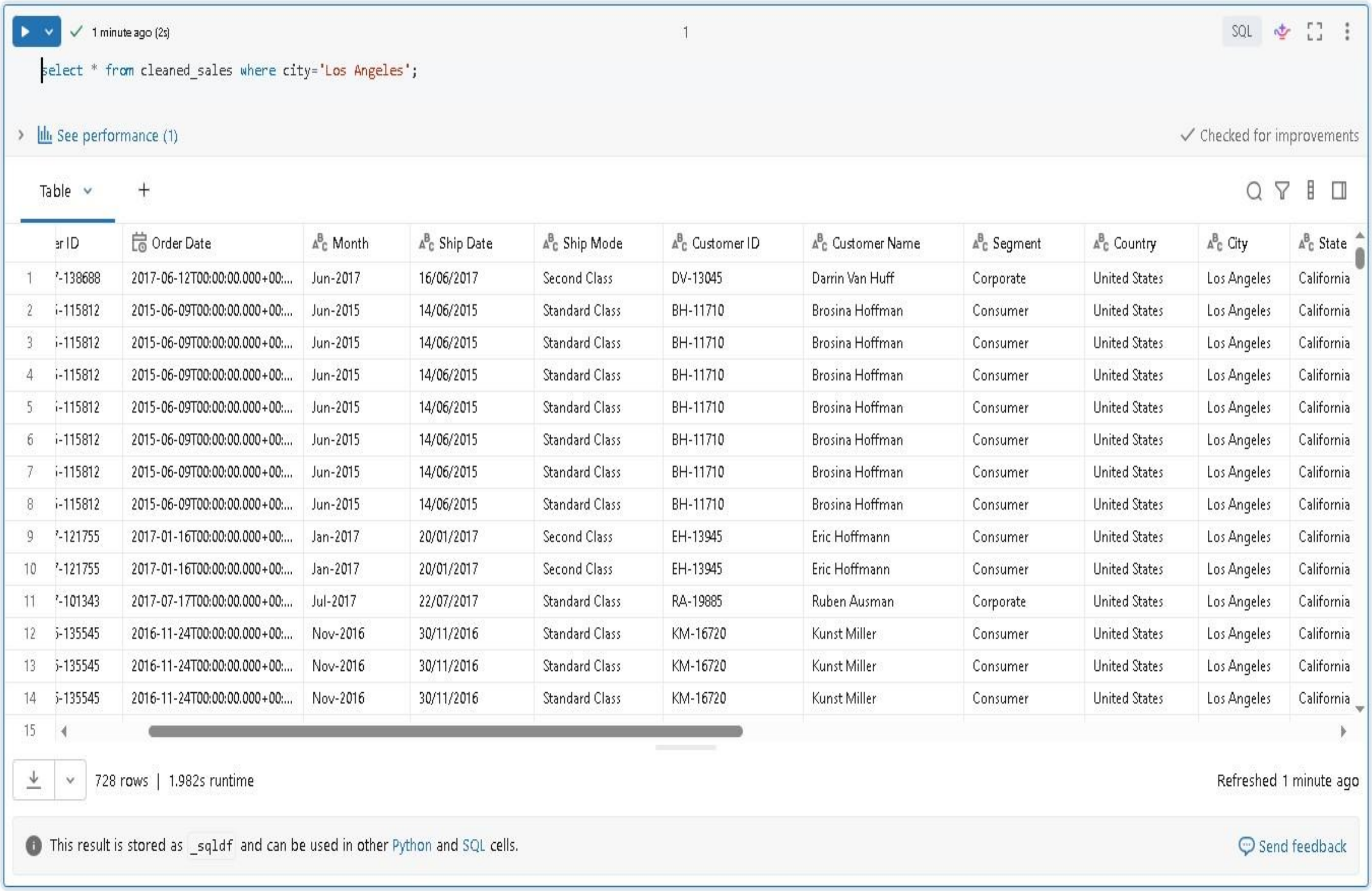
See performance (1)

Table

	category	total_sales
1	Technology	827250.843000000004540...
2	Furniture	727943.375699999999458...
3	Office Supplies	705422.333999999999473...

2.Show sales only for los angles

```sql
select * from cleaned_sales where city='Los Angeles';
```



	er ID	Order Date	Month	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State
1	-138688	2017-06-12T00:00:00.000+00:...	Jun-2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California
2	-115812	2015-06-09T00:00:00.000+00:...	Jun-2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California
3	-115812	2015-06-09T00:00:00.000+00:...	Jun-2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California
4	-115812	2015-06-09T00:00:00.000+00:...	Jun-2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California
5	-115812	2015-06-09T00:00:00.000+00:...	Jun-2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California
6	-115812	2015-06-09T00:00:00.000+00:...	Jun-2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California
7	-115812	2015-06-09T00:00:00.000+00:...	Jun-2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California
8	-115812	2015-06-09T00:00:00.000+00:...	Jun-2015	14/06/2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California
9	-121755	2017-01-16T00:00:00.000+00:...	Jan-2017	20/01/2017	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California
10	-121755	2017-01-16T00:00:00.000+00:...	Jan-2017	20/01/2017	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California
11	-101343	2017-07-17T00:00:00.000+00:...	Jul-2017	22/07/2017	Standard Class	RA-19885	Ruben Ausman	Corporate	United States	Los Angeles	California
12	-135545	2016-11-24T00:00:00.000+00:...	Nov-2016	30/11/2016	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California
13	-135545	2016-11-24T00:00:00.000+00:...	Nov-2016	30/11/2016	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California
14	-135545	2016-11-24T00:00:00.000+00:...	Nov-2016	30/11/2016	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California
15											

3.Show total sales by each city

```sql
SELECT
    city,
    SUM(sales) AS total_sales
FROM
    cleaned_sales
GROUP BY
    city
ORDER BY
    total_sales DESC;
```

Just now (2s)

```sql
SELECT
    city,
    SUM(sales) AS total_sales
FROM
    cleaned_sales
GROUP BY
    city
ORDER BY
    total_sales DESC;
```

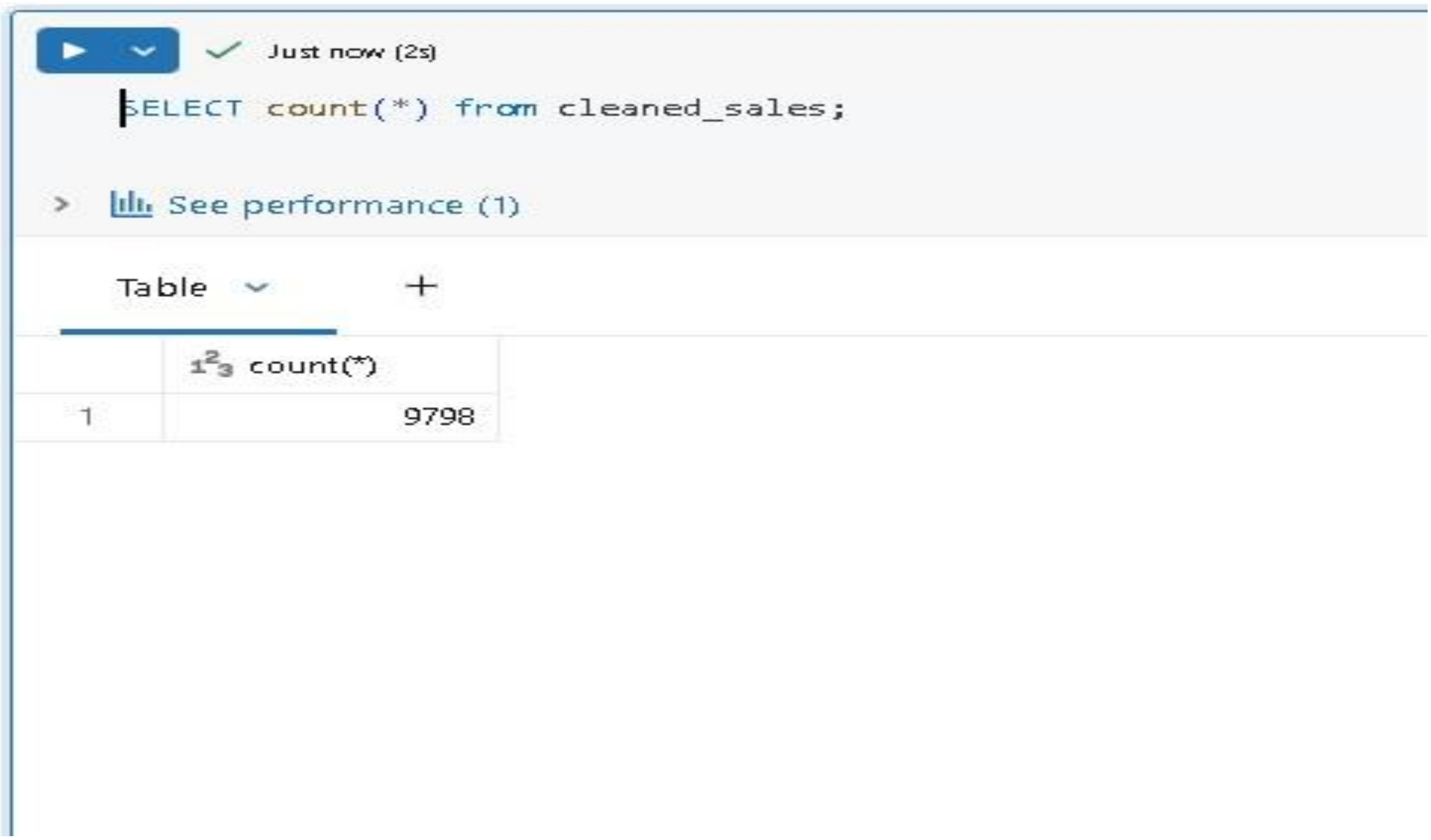
See performance (1)

Table

	city	total_sales
1	New York City	252462.54700000000120...
2	Los Angeles	173420.18099999999999...
3	Seattle	116106.32200000000380...
4	San Francisco	109041.12000000000017...
5	Philadelphia	108841.74900000000001151...
6	Houston	63956.14280000000037500
7	Chicago	47820.13300000000014697
8	San Diego	47521.02900000000013690
9	Jacksonville	44713.18299999999998270
10	Detroit	42446.94400000000011660
11	Springfield	41827.81000000000003530
12	Columbus	38662.56300000000020080
13	Newark	28448.04899999999995320

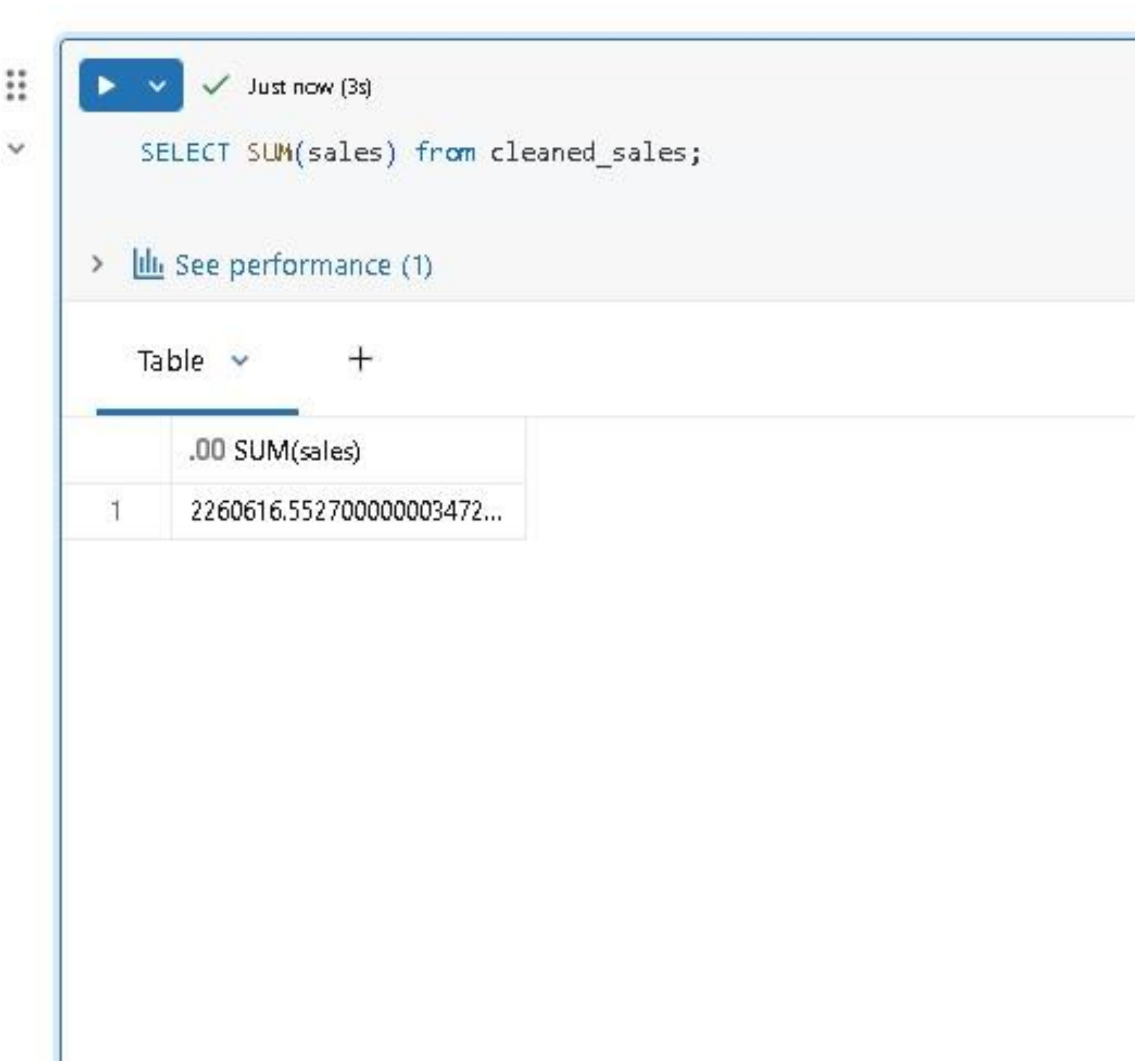
4. How many sales transactions are there?

```
SELECT count(*) from cleaned_sales;
```



5.What is the total revenue?

```sql
SELECT SUM(sales) from cleaned_sales;
```



Insights:

1. Sales by Category

“Technology products generated the highest sales among all categories, indicating strong customer demand in this segment”
“Office Supplies contributed lower sales compared to other categories, suggesting an opportunity for promotional offers”

2.Sales by City

“New York City recorded the highest sales, making it the strongest performing market”

“City Jackson had lower sales, which may require targeted marketing strategies to improve performance”